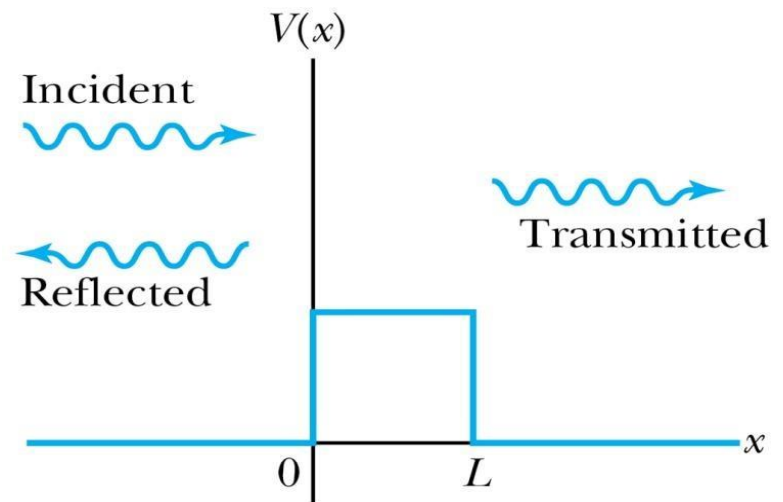
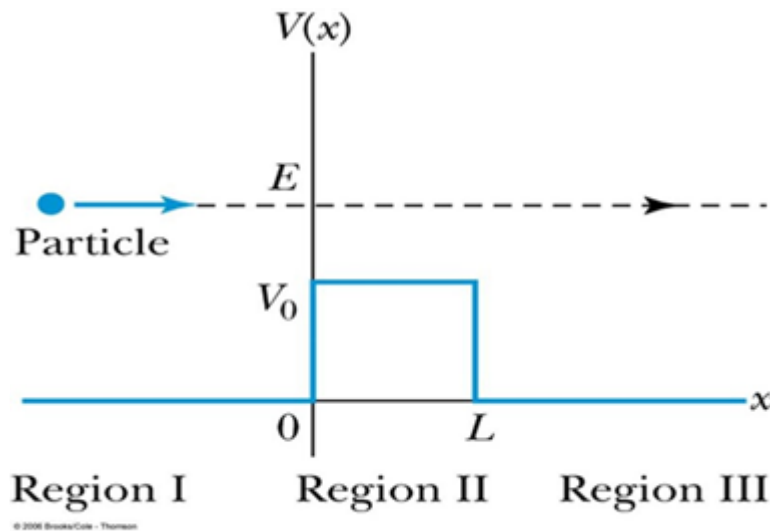


Barrier and Tunneling

- Consider a particle of energy E approaching a potential barrier of height V_0 and the potential everywhere else is zero.
- We will first consider the case when the energy is greater than the potential barrier.
- In regions I and III the wave numbers are: $k_I = k_{III} = \frac{\sqrt{2mE}}{\hbar}$
- In the barrier region we have $k_{II} = \frac{\sqrt{2m(E - V_0)}}{\hbar}$ where $V = V_0$



- The wave function will consist of an incident wave, a reflected wave, and a transmitted wave.
- The potentials and the Schrödinger wave equation for the three regions are as follows:

Region I ($x < 0$)	$V = 0$	$\frac{d^2\psi_I}{dx^2} + \frac{2m}{\hbar^2} E\psi_I = 0$
Region II ($0 < x < L$)	$V = V_0$	$\frac{d^2\psi_{II}}{dx^2} + \frac{2m}{\hbar^2} (E - V_0)\psi_{II} = 0$
Region III ($x > L$)	$V = 0$	$\frac{d^2\psi_{III}}{dx^2} + \frac{2m}{\hbar^2} E\psi_{III} = 0$

- The corresponding solutions are:

Region I ($x < 0$)	$\psi_I = Ae^{ik_I x} + Be^{-ik_I x}$
Region II ($0 < x < L$)	$\psi_{II} = Ce^{ik_{II} x} + De^{-ik_{II} x}$
Region III ($x > L$)	$\psi_{III} = Fe^{ik_I x} + Ge^{-ik_I x}$

Incident wave	$\psi_I(\text{incident}) = Ae^{ik_I x}$
Reflected wave	$\psi_I(\text{reflected}) = Be^{-ik_I x}$
Transmitted wave	$\psi_{III}(\text{transmitted}) = Fe^{ik_I x}$

Probability of Reflection and Transmission

- The probability of the particles being reflected R or transmitted T is:

$$R = \frac{|\psi_I(\text{reflected})|^2}{|\psi_I(\text{incident})|^2} = \frac{B^* B}{A^* A}$$

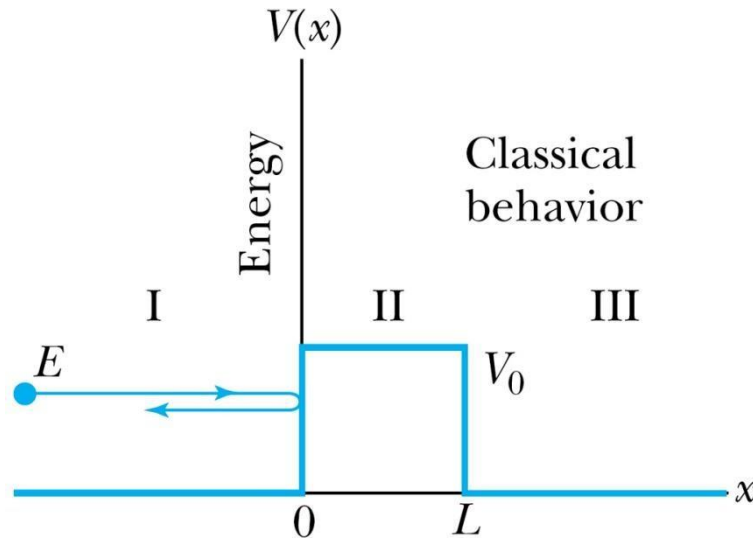
$$T = \frac{|\psi_{III}(\text{transmitted})|^2}{|\psi_I(\text{incident})|^2} = \frac{F^* F}{A^* A}$$

- The maximum kinetic energy of the photoelectrons depends on the value of the light frequency and not on the intensity.
- Because the particles must be either reflected or transmitted, $R + T = 1$.
- By applying the boundary conditions $x \rightarrow \pm\infty$, $x = 0$, and $x = L$, we arrive at the transmission probability:

$$T = \left[1 + \frac{V_0^2 \sin^2(k_{II} L)}{4E(E - V_0)} \right]^{-1}$$

- Notice that there is a situation in which the transmission probability is 1.

- ❖ Now we consider the situation particle does not have enough energy to surmount the potential barrier, $E < V_0$.



- The quantum mechanical result, however, is one of the most remarkable features of modern physics, and there is ample experimental proof of its existence. There is a small, but finite, probability that the particle can penetrate the barrier and even emerge on the other side.
- The wave function in region II becomes $\psi_{II} = Ce^{\kappa x} + De^{-\kappa x}$ where $\kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$
- The transmission probability that describes the phenomenon of **tunneling** is $T = \left[1 + \frac{V_0^2 \sinh^2(\kappa L)}{4E(V_0 - E)} \right]^{-1}$

Some Applications of Quantum Tunneling

1. Scanning tunneling Microscope
2. Alpha particle decay
3. tunnel Diodes
4. Tunnel Field Effect Transistors
5. Nuclear Fusion in Stars
6. Tunnel Junctions

Types of Electron Microscopy

Scanning Electron Microscope

